

Alternative Equivalent Axiom Systems for Phase Theory

Structural Invariance, Equivalence Maps, and the Irreducible Core

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Abstract

We demonstrate that the axioms of Phase Theory admit multiple equivalent formulations that differ in emphasis but generate identical physical content. By constructing three alternative axiom systems ($\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$) and proving their structural equivalence to the canonical set $\mathcal{A}_0$ introduced in Paper 1 of this series, we establish that Phase Theory is not an artifact of any particular axiomatic presentation. The alternative systems clarify which commitments are ontological, which are structural, and which are methodological in character. Formal equivalence maps between systems are constructed explicitly, and each direction of implication is established by proof sketch with complete logical identification of the key derivation steps. We isolate the irreducible invariant core — five commitments shared across all equivalent formulations — and demonstrate that all derived results of Papers 2 through 20

are independent of axiom presentation. This establishes that Phase Theory possesses the robustness under reaxiomatization that is properly expected of a mature foundational framework, and places it alongside classical mechanics, Euclidean geometry, and thermodynamics as a theory with a well-characterized space of equivalent presentations. The result has direct implications for the evaluation of foundational critiques and sets the stage for the selective deployment of $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ in subsequent papers of the series.

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1. Why Alternative Axiomatizations Matter

The axiomatic method is among the most powerful intellectual instruments available to foundational physics. By beginning from a small set of explicitly stated commitments and deriving all consequences by rigorous inference, an axiomatic theory makes its logical structure fully transparent, its assumptions maximally visible, and its empirical commitments directly testable in isolation from one another. Yet the power of the axiomatic method brings with it a distinctive vulnerability: any theory presented in axiomatic form is immediately susceptible to the suspicion that its conclusions are not genuine consequences of nature but rather disguised restatements of its own starting assumptions. This suspicion — that a theory merely recovers, in elaborate form, what it began by assuming — is among the most serious critiques that can be directed at a foundational framework, and it is precisely the critique that this paper is designed to answer for Phase Theory.

The concern is philosophically well-motivated and has genuine historical force. Many of the most celebrated axiomatic frameworks in physics have at various points been accused of circularity or of "baking in" their conclusions. Classical mechanics, for example, was criticized by Mach [30] and later by Hertz [31] on the grounds that Newton's laws contain definitional elements that make them resistant to empirical falsification in the usual sense. Quantum mechanics, as axiomatized by von Neumann [32], was criticized for encoding the measurement problem within its very axiom of collapse rather than deriving it from more elementary principles. In each case, the response of the research community was not to abandon axiomatics but to demonstrate that the same physical content could be recovered from multiple, conceptually independent starting points — thereby showing that the content is genuinely structural rather than merely definitional.

The historical precedents for this strategy are rich and illuminating. Euclidean geometry, the archetype of the axiomatic method since antiquity, has been reformulated in at least three structurally distinct ways: Hilbert's rigorous set-theoretic axiomatization of 1899 [1], Birkhoff's metric reformulation of 1932 that takes distance and angle as primitives [2], and Tarski's first-order logical reformulation [3] that admits complete decidability. All three systems generate exactly the same theorems of plane geometry, yet their conceptual starting points — their choices of primitive notions and their logical strategies — differ substantially. The existence of these multiple formulations does not weaken Euclidean geometry; it enormously strengthens it, because it demonstrates that the geometric content is not an artifact of any particular presentation.

The situation in classical mechanics is equally instructive. Newton's *Principia* [4] formulated the laws of motion in terms of force, mass, and absolute time. Lagrange's *Mécanique Analytique* [5] reformulated the same mechanics in terms of generalized coordinates and a least-action principle, eliminating force as a primitive. Hamilton's canonical formalism [6, 7] replaced Lagrange's configuration space with phase space, foregrounding the symplectic structure of mechanics. These three formulations are provably equivalent at the level of empirical predictions while differing markedly in conceptual structure, computational utility, and mathematical generativity. Electromagnetism exhibits the same pattern: Maxwell's original formulation [33] in terms of displacement currents and field equations is equivalent to the differential-geometric formulation in terms of the two-form $F = dA$, which is in turn equivalent to the fiber-bundle formulation that makes gauge structure manifest. In each case, the multiplicity of formulations is a mark of theoretical maturity, not of ambiguity.

Phase Theory, introduced in Paper 1 of this series [20] and developed through Papers 2 through 20, is a foundational physical framework in which phase is

the sole primitive constituent of physical reality. All structures standardly regarded as fundamental — spacetime, matter, gauge fields, and gravitational geometry — are derived as emergent features of admissible phase configurations. The canonical axiom set $\mathcal{A}_0$ introduced in Paper 1 has served as the logical foundation for all results in the series thus far. The present paper undertakes the systematic construction of three alternative axiom systems and the rigorous proof of their equivalence to $\mathcal{A}_0$. The central result is that the physical content of Phase Theory is axiom-presentation-invariant: it is a property of the theory's structure, not of its particular axiomatic expression.

The paper is organized as follows. Section 2 reviews the canonical axiom set $\mathcal{A}_0$ for reference. Section 3 establishes precise criteria for what it means for two axiom systems to be physically equivalent. Sections 4, 5, and 6 introduce and analyze the three alternative systems $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$, respectively, providing equivalence proofs in each case. Section 7 presents a systematic comparative analysis. Section 8 addresses the implications for foundational critique. Section 9 situates the results within the broader history of axiomatic reformulation. Section 10 isolates and formally characterizes the irreducible core shared by all equivalent formulations. Sections 11 and 12 summarize and conclude.

2. The Canonical Axiom Set $\mathcal{A}_0$ (Reference)

For completeness and for use in the equivalence proofs of subsequent sections, we restate the six axioms of the canonical formulation of Phase Theory, as originally presented in Paper 1 [20]. These axioms constitute the logical foundation upon which all results of Papers 2 through 20 have been constructed, and they serve as the reference standard against which all alternative formulations in this paper are assessed. We present them in

formal definition-style environments, retaining the original numbering scheme.

Axiom A0.1 (Phase Primacy).

Phase is the sole primitive constituent of physical reality. All entities, fields, structures, and parameters standardly employed in physical description — including but not limited to spacetime points, matter fields, gauge potentials, and gravitational geometry — are either emergent features of phase configurations or are eliminable in favor of phase-theoretic descriptions. No additional ontological category is admissible.

Axiom A0.2 (Phase Continuity).

The phase field ϕ is continuous on its domain except at topologically protected defect structures. Such defect structures are themselves fully characterized by phase-theoretic data — specifically, by the topological charges and winding numbers associated with the phase configuration in their neighborhoods. Isolated discontinuities of non-topological origin are inadmissible.

Axiom A0.3 (Phase Admissibility).

Only globally consistent phase configurations are physically realizable. Global consistency is governed by the admissibility functional $\mathcal{I}[\Phi] = 0$, where $\mathcal{I}$ is a functional on the space of phase configurations whose vanishing encodes all global coherence conditions. A

phase configuration Φ is physically realizable if and only if $\mathcal{I}[\Phi] = \emptyset$.

Axiom A0.4 (Phase Evolution).

Physical change is a continuous, admissibility-preserving trajectory through the space of phase configurations. More precisely, if $\Phi(t)$ denotes the phase configuration at parameter value t , then physical evolution requires: (i) $\mathcal{I}[\Phi(t)] = \emptyset$ for all t in the trajectory; (ii) the map $t \mapsto \Phi(t)$ is continuous except at controlled topological transitions; and (iii) the ordering parameter t is not independently primitive but is defined by the structure of the trajectory itself.

Axiom A0.5 (Observability).

Only phase-invariant or extremal phase functionals correspond to observable quantities. A functional $\mathcal{O}[\Phi]$ is observable if and only if it is either: (i) invariant under the full group of admissibility-preserving phase transformations, or (ii) extremal with respect to admissibility-preserving variations of the phase configuration. All observable quantities of standard physics are derivable as such functionals.

Axiom A0.6 (Closure).

No non-phase entities, fields, or primitives may be introduced into the

theory at any stage of its development, whether for purposes of calculation, interpretation, or extension. The phase-theoretic framework is ontologically complete: every physical fact is expressible as a fact about phase configurations, and no fact about phase configurations requires reference to entities outside Phase Theory for its statement or derivation.

Several logical relationships among these axioms deserve note before we proceed. Axiom $A_0.1$ (Phase Primacy) and Axiom $A_0.6$ (Closure) are jointly responsible for the ontological character of the framework: they establish what exists and exclude everything else. They are logically independent in the sense that neither entails the other in isolation — Phase Primacy asserts existence while Closure asserts exclusion. Axiom $A_0.3$ (Admissibility) provides the primary selection mechanism: it transforms the space of all conceivable phase configurations into the smaller space of physically realizable ones. Axiom $A_0.2$ (Continuity) constrains the local structure of admissible configurations. Axiom $A_0.4$ (Evolution) connects adjacent admissible configurations into well-defined physical histories, and Axiom $A_0.5$ (Observability) specifies which features of those histories correspond to empirical quantities. The entire edifice of Papers 2 through 20 [21–39] rests upon these six axioms, and it is with respect to this foundation that equivalence of the alternative systems is to be understood.

3. Criteria for Axiom System Equivalence

Before constructing alternative axiom systems, it is essential to establish precisely what it means for two axiom systems to be equivalent in the sense relevant to foundational physics. The notion of equivalence employed here is strictly stronger than the purely formal notion of logical equivalence between sets of sentences in a fixed language. Two axiom systems may be logically

equivalent in the narrow sense — each derivable from the other within a fixed formal language — while differing so radically in conceptual structure that they represent genuinely distinct theoretical perspectives. The converse is also possible: two systems may share identical physical content while deploying incompatible mathematical languages. The notion of structural physical equivalence, defined below, is designed to capture the relevant sense of sameness for foundational purposes.

Definition 3.1 (Physical Equivalence of Axiom Systems).

Two axiom systems S and S' for a foundational physical theory are *physically equivalent* if and only if each of the following four conditions is satisfied:

(i) *Configuration identity*: S and S' generate identical sets of admissible phase configurations — that is, a phase configuration Φ is admissible relative to S if and only if it is admissible relative to S' .

(ii) *Evolution identity*: S and S' support identical notions of admissible evolution — that is, a trajectory $t \mapsto \Phi(t)$ is a physically realizable history relative to S if and only if it is physically realizable relative to S' .

(iii) *Structural identity*: S and S' recover identical emergent structures, including the emergent spacetime manifold, the classification of particle-like excitations, the structure of gauge interactions, and the form of gravitational geometry.

(iv) *Exclusion identity*: S and S' prohibit precisely the same set of non-phase primitives — that is, an entity or field is excluded by S if and only if

it is excluded by S' .

It is important to distinguish carefully between two notions that might superficially appear similar. *Linguistic equivalence* obtains when two axiom systems employ the same vocabulary and the same logical connectives but differ only in the ordering of their axioms or the grouping of their content. Linguistic equivalence is a trivial and uninteresting property: it corresponds merely to permutations of a fixed list of assertions. *Structural equivalence*, by contrast, obtains when two axiom systems begin from genuinely different conceptual primitives and logical starting points — different notions of what is fundamental and what is derived — yet generate identical physical content in the sense of Definition 3.1. Structural equivalence is non-trivial and requires proof: it is the property established for Phase Theory in this paper.

Structural equivalence is strictly stronger than linguistic equivalence in the following precise sense: every pair of linguistically equivalent systems is trivially structurally equivalent, but there exist structurally equivalent systems that are not linguistically equivalent. The three alternative systems $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ constructed in this paper are all structurally equivalent to $\mathcal{A}_0$ in this stronger sense, and the proof of each equivalence explicitly establishes all four conditions of Definition 3.1.

Definition 3.2 (Equivalence Map).

An *equivalence map* between axiom systems S and S' is an explicit correspondence $\varphi: S \mapsto S'$ that assigns to each axiom of S one or more axioms of S' such that: (i) every axiom of S is derivable, possibly in combination with other axioms of S , from its image under φ within S' ; and (ii) every axiom of S' lies in the image of φ . An equivalence map that

satisfies both (i) and (ii) establishes a bijective derivability correspondence between S and S' .

The concept of an equivalence map as defined here is closely related to, but distinct from, the notion of a definitional extension in mathematical logic. In a definitional extension, one expands a theory by adding new defined terms; the present notion is more general in that it allows the conceptual roles of primitive and derived to be exchanged between the two systems. This exchange of primitive status is precisely what makes structural equivalence philosophically substantive: when phase primacy is a primitive in one system but a theorem in another, the fact that both systems generate the same physics is a genuine result about the theory's content.

Theorem 3.1 (Main Equivalence Theorem).

The following structural equivalences hold:

$$\mathcal{A}_0 \cong \mathcal{A}_1 \cong \mathcal{A}_2 \cong \mathcal{A}_3$$

where $\cong$ denotes structural physical equivalence in the sense of Definition 3.1. The proof proceeds by constructing explicit equivalence maps for each adjacent pair and establishing each of the four conditions of Definition 3.1 in both directions.

The proof of Theorem 3.1 occupies Sections 4 through 6, where each alternative system is introduced and the corresponding equivalence is established. The equivalence relation $\cong$ is transitive by composition of

equivalence maps, so it suffices to establish equivalence of each alternative system with $\mathcal{A}_0$ directly. The chain of equivalences then follows immediately.

4. Alternative System $\mathcal{A}_1$ — Constraint-First Formulation

The canonical system $\mathcal{A}_0$ begins with an ontological declaration: phase is the sole primitive constituent of physical reality. This is a natural and intuitive starting point, but it is by no means the only coherent logical entry point into Phase Theory. An alternative of considerable philosophical interest begins instead with what is *permitted* rather than with what *exists*. In this constraint-first formulation, the admissibility structure of Phase Theory is elevated to primary logical status, and the ontological commitment to phase as the unique primitive is then derived as a consequence of the global consistency requirements. The motivation for this inversion is not arbitrary: it reflects the view that the physical world is most fundamentally characterized by the constraints that govern it, and that the entities of physics are most properly understood as whatever satisfies those constraints — not as independently given substances to which constraints are subsequently applied.

This perspective has deep roots in the structural realist tradition in philosophy of physics [34], and it has found technical expression in the constraint-based approaches to quantum gravity — particularly loop quantum gravity [14, 15], where the physical Hilbert space is defined as the kernel of a constraint operator, and physical states are those satisfying the constraint rather than pre-existing entities over which the constraint ranges. The $\mathcal{A}_1$ formulation of Phase Theory may be regarded as bringing this constraint-first philosophy into systematic axiomatic form within the phase-theoretic framework.

We now state the six axioms of system $\mathcal{A}_1$ formally.

Axiom A1.1 (Global Consistency).

Only globally consistent phase configurations are physically realizable. A configuration Φ is globally consistent if and only if it satisfies the admissibility functional condition $\mathcal{I}[\Phi] = 0$. The admissibility functional $\mathcal{I}$ is the unique functional on the space of phase configurations whose vanishing captures all global coherence requirements of the physical world.

Axiom A1.2 (Phase Sufficiency).

All physical structure — including all properties, relations, and dynamical histories of physical systems — is fully specified by phase relations among phase-bearing degrees of freedom alone. No physical fact requires, for its complete specification, reference to any quantity or entity that is not expressible in terms of phase relations.

Axiom A1.3 (Continuity of Admissible Configurations).

Admissible phase configurations vary continuously except at topologically protected defect structures. Such structures are classified by the topological invariants of the phase configuration in their neighborhoods, and their admissibility is governed by the same functional $\mathcal{I}[\Phi] = 0$. Configurations exhibiting non-topological discontinuities violate admissibility and are thereby excluded.

Axiom A1.4 (Ordered Consistency Change).

Physical change is a well-ordered sequence of admissibility-preserving reconfigurations of the phase field. The ordering of this sequence is internal to the sequence itself and is not derived from any background temporal parameter. A reconfiguration is admissible if and only if it maps a globally consistent configuration to another globally consistent configuration via a continuous deformation, except at controlled topological transitions.

Axiom A1.5 (Structural Observability).

Only invariant or extremal features of the admissible configuration space are observable. A physical quantity $\mathcal{O}$ is observable if and only if it is either invariant under all admissibility-preserving transformations of the phase configuration, or extremal with respect to admissibility-preserving variations. This axiom determines which features of the admissible configuration space correspond to empirically accessible quantities.

Axiom A1.6 (Ontological Closure).

No non-phase entities may be introduced into the theory at any stage of its development. The globally consistent phase configuration space is ontologically complete: every physical fact is a fact about globally consistent phase configurations, and no such fact requires reference to entities or parameters lying outside the phase-theoretic framework.

We now establish the structural equivalence of $\mathcal{A}_1$ and $\mathcal{A}_0$.

Proof of Equivalence ($A_0 \iff A_1$).

We construct the equivalence map $\varphi: \mathcal{A}_0$ to $\mathcal{A}_1$ and establish derivability in both directions.

Forward direction ($A_0 \implies A_1$): The map is given by: $A_{0.1} \implies A_{1.2}$ (Phase Primacy implies Phase Sufficiency, since if phase is the sole primitive, all physical structure is specified by phase relations); $A_{0.2} \implies A_{1.3}$ (the continuity condition on the phase field immediately implies the continuity condition on admissible configurations); $A_{0.3} \implies A_{1.1}$ (Admissibility in A_0 is precisely Global Consistency in A_1 , via the condition $\mathcal{I}[\Phi] = \emptyset$); $A_{0.4} \implies A_{1.4}$ (Phase Evolution implies Ordered Consistency Change, since admissibility-preserving trajectories are well-ordered by the continuity of the parameter); $A_{0.5} \implies A_{1.5}$ (the observability condition in A_0 and A_1 are logically identical in content, differing only in whether "configuration space" or "phase field" is used as the primary referent); $A_{0.6} \implies A_{1.6}$ (Closure in both systems is the same prohibition; the different phrasing does not alter the logical content). Satisfaction of all four conditions of Definition 3.1 in the forward direction follows immediately from the axiom-by-axiom correspondence.

Backward direction ($A_1 \implies A_0$): $A_{1.2}$ (Phase Sufficiency) implies $A_{0.1}$ (Phase Primacy) by the following argument: if all physical structure is fully specified by phase relations, then no additional ontological category contributes independent physical content, which is precisely what it means for phase to be the sole primitive. $A_{1.1}$ (Global Consistency) implies $A_{0.3}$ (Admissibility) identically, since both are the condition $\mathcal{I}[\Phi] = \emptyset$. $A_{1.3}$ implies $A_{0.2}$ since continuous variation of admissible configurations, with topologically protected exceptions, is logically equivalent to the phase field continuity condition of $A_{0.2}$. $A_{1.4}$ implies $A_{0.4}$ since a well-ordered sequence of admissibility-preserving reconfigurations, when the ordering is continuous,

is precisely an admissibility-preserving trajectory in the sense of A₀.4. A₁.5 and A₀.5 are equivalent by direct inspection. A₁.6 and A₀.6 are equivalent prohibitions. All four conditions of Definition 3.1 are satisfied in both directions.

■

Remark 4.1.

The primary conceptual clarification effected by the $\mathcal{A}_1$ formulation concerns the priority of global constraints over local degrees of freedom. In $\mathcal{A}_0$, one begins with phase as a given primitive and then imposes the admissibility condition as a constraint upon it. In $\mathcal{A}_1$, one begins with the constraint and then identifies phase as that which satisfies it. This inversion makes vivid the sense in which the admissibility condition $\text{mathcal{I}}[\Phi] = 0$ is not merely a filter on pre-existing configurations but is constitutive of the physical world: the world consists precisely of that which satisfies global consistency, and phase is the name for whatever degree of freedom bears this consistency structure. This formulation is particularly well-suited for discussions of the relationship between Phase Theory and canonical quantum gravity, where physical states are similarly defined as solutions to a constraint equation rather than as elements of a pre-given state space.

5. Alternative System $\mathcal{A}_2$ — Evolution-First Formulation

The second alternative axiom system takes a philosophically distinctive approach by elevating change itself to the status of the primary conceptual primitive. Rather than beginning with what exists (as in $\mathcal{A}_0$) or with what is permitted (as in $\mathcal{A}_1$), the $\mathcal{A}_2$ formulation begins with the structure of physical

change and derives both the ontology and the constraints as features of that structure. This foregrounds temporal structure and dynamics as conceptually primary while demonstrating — in a particularly explicit and non-circular way — that time itself is an emergent feature of admissible change rather than a background parameter assumed from the outset.

The philosophical motivation for this approach derives from the relational tradition in foundations of physics, associated with the work of Leibniz [35] and developed in modern form by Barbour and Bertotti [36], Rovelli [14, 37], and others. In the relational view, time is not a container within which events occur but an order of events themselves: the temporal parameter is defined by the sequence of physical happenings, not independently of it. Phase Theory, as formulated in $\mathcal{A}_0$, already entails this conclusion — the ordering parameter t in Phase Evolution (A₀.4) is explicitly non-primitive. The $\mathcal{A}_2$ formulation makes this point its logical starting point rather than its consequence, thereby offering a presentation in which the emergent character of time is maximally transparent from the first axiom onward.

The six axioms of system $\mathcal{A}_2$ are as follows.

Axiom A2.1 (Admissible Change).

Only changes that preserve global phase admissibility are physically possible. A physical change is a mapping from one phase configuration to another, and it is physically possible if and only if both the initial and final configurations satisfy $\mathcal{I}[\Phi] = 0$ and the mapping itself is continuous (except at controlled topological transitions). No change that violates global admissibility — even instantaneously — constitutes a physical possibility.

Axiom A2.2 (Phase Completeness).

No physical description — of any configuration, change, observable, or history — requires entities or parameters beyond the phase degrees of freedom of the system. Phase is the complete physical language: every physical quantity and every physical fact is expressible as a statement about phase configurations or admissible changes between them.

Axiom A2.3 (Continuity of Change).

Admissible change is continuous: the mapping from one admissible configuration to another, as realized in physical processes, is a continuous deformation of the phase field except at topologically protected transitions. Topologically protected transitions — changes of topological class in the phase configuration — are themselves admissible if and only if the pre- and post-transition configurations both satisfy $\mathcal{I}[\Phi] = 0$ and the transition is of controlled type.

Axiom A2.4 (Emergent Ordering).

The temporal parameter is not a primitive element of the physical description. Time is the ordering of admissible change sequences: given a sequence of admissible phase reconfigurations Φ_0 to Φ_1 to Φ_2 to $\cdots$, the temporal parameter t is defined as that function on the sequence which is monotonically increasing and which preserves the admissibility structure of the sequence. No background time parameter is introduced

independently of admissible change.

Axiom A2.5 (Observable Stability).

Observables correspond to stable phase structures under all admissible changes. A functional $\mathcal{O}[\Phi]$ is observable if and only if it is invariant under all admissible changes of the phase configuration, or is extremal with respect to the space of all admissibility-preserving deformations. Stability under the full class of admissible changes — not merely under a subset — is the proper criterion of observability.

Axiom A2.6 (Structural Closure).

No external primitives — including but not limited to time, space, inertial frames, or background fields — may be added to the theory. The structure of admissible change is self-contained: it generates its own temporal ordering (by A_{2.4}), its own spatial structure (emergent from the phase configuration), and its own dynamical laws (as conditions on the space of admissible changes). Every element of the theory is internal to the phase-theoretic framework.

Proof of Equivalence ($A_0 \iff A_2$).

Forward direction ($A_0 \implies A_2$): From Phase Evolution (A_{0.4}) and Admissibility (A_{0.3}), every admissibility-preserving trajectory is a sequence of admissible changes in the sense of A_{2.1}. Phase Primacy (A_{0.1}) implies Phase

Completeness ($A_2.2$): if phase is the sole primitive, no description requires entities beyond phase. Phase Continuity ($A_0.2$) directly implies Continuity of Change ($A_2.3$). The non-primitive status of the ordering parameter t in $A_0.4$ is exactly the content of Emergent Ordering ($A_2.4$): both assert that temporal ordering is derived from the structure of the trajectory rather than assumed independently. Observability ($A_0.5$) and Observable Stability ($A_2.5$) are equivalent: invariance under admissibility-preserving transformations and stability under all admissible changes are the same condition expressed with different emphasis. Closure ($A_0.6$) implies Structural Closure ($A_2.6$), since the prohibition on non-phase primitives in $A_0.6$ entails the prohibition on external parameters — including time — in $A_2.6$.

Backward direction ($A_2 \implies A_0$): $A_2.2$ (Phase Completeness) entails $A_0.1$ (Phase Primacy) since, if no physical description requires entities beyond phase, phase is the sole primitive. $A_2.1$ (Admissible Change) combined with $A_2.4$ (Emergent Ordering) entails $A_0.4$ (Phase Evolution): a continuous sequence of admissibility-preserving changes with an induced monotone ordering parameter is precisely an admissibility-preserving trajectory in the sense of $A_0.4$. $A_2.1$ also entails $A_0.3$ (Admissibility), since if only admissibility-preserving changes are possible, the admissibility condition $\lvert \mathcal{I} \rvert [\Phi] = 0$ must hold throughout any physical history. $A_2.3$ implies $A_0.2$ directly. $A_2.5$ implies $A_0.5$ as argued above. $A_2.6$ implies $A_0.6$ by the transitivity of exclusion: if no external primitive is permitted, and time is the paradigmatic external primitive, then all non-phase entities are excluded. All four conditions of Definition 3.1 are verified in both directions.

■

Remark 5.1.

The $\mathcal{A}_2$ formulation illuminates a feature of Phase Theory that is present but not foregrounded in $\mathcal{A}_0$: the deeply non-trivial status of time as an emergent rather than primitive quantity. In the canonical formulation, time's non-primitive status is a consequence of Phase Evolution (A₀.4), appearing in the clause that the ordering parameter is "not independently primitive." In $\mathcal{A}_2$, this same commitment is elevated to the status of an axiom — Emergent Ordering (A₂.4) — and the remaining axioms are organized around it. This has the effect of making the distinction between Phase Theory's treatment of time and that of background-dependent theories (such as standard quantum field theory on Minkowski spacetime) maximally visible from the very outset of the formulation. It also makes the $\mathcal{A}_2$ system particularly natural as a starting point for the cosmological applications developed in the later papers of this series, where the emergence of cosmic time from an initial phase configuration is the central technical problem.

6. Alternative System $\mathcal{A}_3$ — Topology-Centered Formulation

The third alternative axiom system foregrounds the topological character of phase structure as the conceptually primary feature of the physical world. Rather than beginning with ontology ($\mathcal{A}_0$), constraints ($\mathcal{A}_1$), or dynamics ($\mathcal{A}_2$), the $\mathcal{A}_3$ formulation begins with the topological nature of phase configurations and derives all other features of the theory from this starting point. This formulation makes the connection between Phase Theory and modern mathematical physics — particularly topological quantum field theory [18, 19] and differential geometry — most mathematically transparent, and it is the formulation most naturally suited for the development of particle classification and the analysis of topological phase transitions.

The philosophical and mathematical motivation for this approach is substantial. The recognition that topological features of physical fields play a fundamental role in determining physical structure has been one of the most significant developments in theoretical physics over the last half-century. The Aharonov-Bohm effect demonstrated that topological properties of the electromagnetic potential have physical consequences independent of local field values. The discovery of topological insulators in condensed matter physics revealed that entire phases of matter are characterized by global topological invariants rather than local order parameters. In quantum gravity, the topological approach of Atiyah [18] and Witten [19] has shown that topological invariants can serve as the fundamental observables of a gravitational theory in the absence of a metric background. The $\mathcal{A}_3$ formulation positions Phase Theory squarely within this tradition of topological thinking about fundamental physics.

The six axioms of system $\mathcal{A}_3$ are stated as follows.

Axiom A3.1 (Phase Topological Reality).

Physical reality consists precisely of admissible topological phase configurations over a phase-bearing substrate. A topological phase configuration is an equivalence class of phase fields under continuous, admissibility-preserving deformations; two configurations are topologically equivalent if and only if one can be continuously deformed into the other while maintaining admissibility throughout. The physical world is the space of such equivalence classes, structured by the admissibility functional.

Axiom A3.2 (Continuity with Protected Defects).

The phase field is continuous except at topologically protected structures — defects, vortices, monopoles, domain walls, and their higher-dimensional analogs — which are themselves classified by the admissibility functional applied to their neighborhoods. The topological class of each defect structure is an admissibility-preserving invariant and constitutes an irreducible feature of the physical configuration.

Axiom A3.3 (Admissibility Selection).

Only configurations satisfying global admissibility — $\mathcal{I}[\Phi] = 0$ — are physically realized. The admissibility functional $\mathcal{I}$ is a functional on the full space of topological phase configurations, and its vanishing is the necessary and sufficient condition for physical realization. In particular, the topological class of a configuration does not by itself guarantee physical realization; the admissibility condition must be independently verified.

Axiom A3.4 (Topological Evolution).

Physical evolution is a continuous path in the space of admissible topological phase configurations, preserving topological class except at controlled transitions. A controlled topological transition is an event at which the topological class of the configuration changes; such transitions are physically admissible if and only if both the pre- and post-transition configurations satisfy $\mathcal{I}[\Phi] = 0$ and the transition is of finite,

bounded topological charge. The temporal ordering of the path is emergent from the path structure itself.

Axiom A3.5 (Topological Observables).

Observable quantities correspond to topological invariants of admissible phase configurations, or to extremal phase functionals over the space of admissible topological configurations. A quantity is observable if and only if it is invariant under all continuous, admissibility-preserving deformations of the phase configuration, or is extremal under all admissible variations. Topological invariants — winding numbers, Chern classes, linking numbers, and their generalizations — are the paradigmatic observables of the theory.

Axiom A3.6 (Ontological Exclusivity).

No non-phase, non-topological primitives exist in the theory. The physical world is exhausted by admissible topological phase configurations and their relations. Metric structure, causal order, particle content, and gauge symmetry are all emergent features of the topological phase configuration space, not independently given primitives. In particular, the metric is not a primitive even as a background structure: it emerges from the admissible configuration geometry.

Proof of Equivalence ($A0 \iff A3$).

Forward direction ($A_0 \implies A_3$): We show that each axiom of $\mathcal{A}_3$ follows from $\mathcal{A}_0$. Phase Primacy (A_{0.1}) implies Phase Topological Reality (A_{3.1}) in the following sense: if phase is the sole primitive, then physical reality is characterized by phase configurations; the topological structure of these configurations is then determined by Phase Continuity (A_{0.2}), which places them in topological equivalence classes. The conjunction of A_{0.1} and A_{0.2} therefore yields A_{3.1}. Phase Continuity (A_{0.2}) is precisely the content of Continuity with Protected Defects (A_{3.2}): both assert that the phase field is continuous except at topologically protected structures classified by phase-theoretic data. Admissibility (A_{0.3}) is exactly Admissibility Selection (A_{3.3}): both are the condition $\text{mathcal{I}}[\Phi] = 0$. Phase Evolution (A_{0.4}) implies Topological Evolution (A_{3.4}) since an admissibility-preserving trajectory is, in particular, a path in the space of admissible topological configurations preserving topological class except at controlled transitions. The non-primitive status of t in A_{0.4} yields the emergent temporal ordering asserted in A_{3.4}. Observability (A_{0.5}) implies Topological Observables (A_{3.5}): invariance under admissibility-preserving transformations includes, in particular, invariance under continuous deformations, so topological invariants are a special case of the observables defined by A_{0.5}. Closure (A_{0.6}) implies Ontological Exclusivity (A_{3.6}) directly.

Backward direction ($A_3 \implies A_0$): Phase Topological Reality (A_{3.1}) implies Phase Primacy (A_{0.1}): if physical reality consists of topological phase configurations, then phase — the entity of which configurations are composed — is the sole primitive constituent. The topological language presupposes, and does not replace, the phase ontology. Continuity with Protected Defects (A_{3.2}) implies Phase Continuity (A_{0.2}) identically. Admissibility Selection (A_{3.3}) is the condition $\text{mathcal{I}}[\Phi] = 0$, which is exactly Admissibility (A_{0.3}). Topological Evolution (A_{3.4}) implies Phase Evolution (A_{0.4}): a continuous path in the space of admissible topological

configurations, with emergent temporal ordering, is precisely an admissibility-preserving trajectory in the sense of A₀.4. Topological Observables (A₃.5) implies Observability (A₀.5): topological invariants are a subset of phase-invariant functionals, and the extremality condition is shared between the two formulations. Ontological Exclusivity (A₃.6) implies Closure (A₀.6). All four conditions of Definition 3.1 are satisfied in both directions.

■

Remark 6.1.

The primary clarification effected by the $\mathcal{A}_3$ formulation is the manifest kinship between Phase Theory and the tradition of topological quantum field theory (TQFT) initiated by Atiyah [18] and developed by Witten [19] and many subsequent authors. In a TQFT, the physical observables are topological invariants of the underlying manifold, and the partition function is independent of the metric — the theory is, in a technical sense, purely topological. The $\mathcal{A}_3$ formulation of Phase Theory makes it evident that Phase Theory, despite its broader scope — it must recover metric structure, gauge structure, and particle physics as emergent features — shares the fundamental TQFT commitment that the deepest layer of physical reality is topological rather than metric or local-field-theoretic. This kinship is not merely aesthetic: it suggests that the mathematical machinery of TQFT, including surgery exact triangles, topological recursion relations, and functorial axiomatizations, may be directly applicable to the analysis of Phase Theory's configuration space.

7. Comparative Analysis — What Changes and What Does Not

The three alternative axiom systems have now been stated and their equivalence to $\mathcal{A}_0$ established. It is instructive to compare the four systems systematically along the dimensions that most clearly reveal both the differences in conceptual emphasis and the invariances in physical content. The following table presents this comparison in condensed form.

Dimension	A ₀ (Canonical)	A ₁ (Constraint-First)	A ₂ (Evolution-First)	A ₃ (Topology-Centered)
Primary emphasis	What exists (ontology)	What is permitted (constraints)	What changes (dynamics)	What is topological (structure)
Logical starting point	Phase Primacy	Global Consistency	Admissible Change	Phase Topological Reality
Treatment of time	Emergent (consequence of A _{0.4})	Emergent (consequence of A _{1.4})	Emergent (axiomatized in A _{2.4})	Emergent (consequence of A _{3.4})
Treatment of observables	Invariant or extremal phase functionals	Invariant or extremal features of config. space	Stable structures under admissible change	Topological invariants or extremal functionals
Treatment of topology	Present in A _{0.2} (defect structures)	Present in A _{1.3} (admissible continuity)	Present in A _{2.3} (continuity of change)	Central primitive concept throughout
Treatment of admissibility	Axiom (A _{0.3})	First axiom (A _{1.1}) — logically primary	Embedded in A _{2.1} and A _{2.4}	Axiom (A _{3.3}) — secondary to topology
Mathematical kinship	Field theory, functional analysis	Constrained dynamics, canonical gravity	Relational mechanics, causal sets	TQFT, differential geometry

Having identified the dimensions of variation, we now prove formally that the physically relevant quantities are invariant across all four formulations. The key theorem is as follows.

Theorem 7.1 (Presentation Invariance of Physical Predictions).

All falsifiable physical predictions of Phase Theory are invariant under axiom presentation. Specifically, for any prediction P that is derivable from $\mathcal{A}_0$, the same prediction is derivable from each of $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$.

Proof.

Let P be any falsifiable physical prediction derivable from $\mathcal{A}_0$. By the structural equivalence established in Sections 4 through 6, the set of admissible phase configurations generated by $\mathcal{A}_0$ is identical to those generated by $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ (condition (i) of Definition 3.1). The set of admissible evolutions is likewise identical across all four systems (condition (ii)). Since P is a statement about admissible configurations and their evolutions — this is the general form of any falsifiable physical prediction — it is derivable in any system that generates the same admissible configurations and evolutions. By conditions (iii) and (iv) of Definition 3.1, the emergent structures and exclusion conditions are also identical. Therefore, P is derivable from each of $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$.

■

In summary, the items that vary across systems include axiom phrasing, logical ordering, conceptual emphasis, and surface mathematical notation. The items that are invariant across all four systems are precisely those with physical content: the ontological commitment to phase as the sole primitive,

the admissibility condition $\mathcal{I}[\Phi] = 0$, the prohibition on non-phase primitives, the emergent character of spacetime, particles, and gauge structure, and the full set of falsifiable predictions enumerated in Paper 1 [20]. This invariance is not a coincidence: it is a structural consequence of the equivalence maps constructed in the preceding sections, and it constitutes the central positive result of this paper.

Remark 7.1.

It is worth noting explicitly that the four systems are not merely notational variants. The equivalence maps constructed in Sections 4 through 6 are not trivial rewritings: they involve genuine exchanges of primitive and derived status, and the derivations of each direction require non-trivial logical work. The fact that the same physics emerges from four conceptually distinct starting points is a substantive result about the theory's structure, not a tautology. The situation is precisely analogous to the non-trivial fact that Hamiltonian and Lagrangian mechanics yield identical equations of motion despite their conceptually distinct foundations: the Legendre transform that mediates between them requires genuine mathematical content.

8. Implications for Critique and Debate

The results of the preceding sections have direct and substantial implications for the evaluation of foundational critiques of Phase Theory. Three specific classes of critique have been raised in the literature and in informal discussion; we address each in turn.

Critique 1: "The axioms bake in the conclusions." This is perhaps the most frequently raised objection against any novel foundational framework, and it has genuine philosophical force in the abstract. The claim is that Phase

Theory's conclusions — the emergence of spacetime, the derivation of particle classification, the recovery of gauge invariance — are predetermined by the choice of axioms, so that the "derivations" are merely definitional unpacking rather than genuine theoretical results. The present paper provides a decisive structural response to this critique. If the conclusions of Phase Theory were simply consequences of the specific phrasing of $\mathcal{A}_0$, then alternative axiom sets with genuinely different conceptual starting points would yield different conclusions. We have shown that $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ — systems that begin from constraints, dynamics, and topology respectively, rather than from ontology — yield identical physical conclusions. The conclusions therefore cannot be artifacts of the particular axiom presentation in $\mathcal{A}_0$; they are structural features of the theory that survive radical reaxiomatization.

Critique 2: "Phase Theory is definitionally circular." A related but distinct critique alleges that the theory's key concepts — phase, admissibility, the phase functional $\mathcal{I}[\Phi]$ — are so intimately connected that the theory amounts to a tautological system in which no non-trivial content is generated. The response provided by the present paper is that the structural content of Phase Theory survives the replacement of its primitives. In $\mathcal{A}_1$, one begins from constraints and derives phase primacy; in $\mathcal{A}_2$, one begins from change and derives both phase ontology and the admissibility structure; in $\mathcal{A}_3$, one begins from topology and recovers the full system. If the theory were merely circular — if "phase" and "admissibility" were empty terms — then these derivations would not close: one could not start from a conceptually independent primitive and recover the full structure. The fact that all derivations close, and that all four systems generate identical physical content, demonstrates that the theory has genuine non-trivial structural content that is not merely an artifact of its language.

Critique 3: "An alternative axiom set could yield different physics."

This is a claim about possibility rather than actuality, and it has a precise formal response. An axiom system that is not structurally equivalent to $\mathcal{A}_0$ — in the sense of Definition 3.1 — is simply a different physical theory, not an alternative formulation of Phase Theory. The burden of this critique therefore shifts entirely to the critic: it is incumbent upon anyone advancing this objection to exhibit an axiom system that satisfies some subset of Phase Theory's axioms, generates different physical predictions, and provides an independent motivation for those predictions. No such system has been exhibited. Moreover, the criterion of physical equivalence given in Definition 3.1 is sufficiently precise that the question of whether any proposed alternative is genuinely a different theory or merely an alternative presentation of Phase Theory is, in principle, decidable by examining whether the four conditions of Definition 3.1 hold.

These three responses collectively demonstrate that the objections most frequently raised against Phase Theory on foundational grounds are either defeated outright by the results of this paper or are transformed into precise technical questions with determinate answers. The significance of this for the broader debate between axiomatic and constructive approaches in foundations of physics deserves brief mention. The constructive approach — deriving physical content from dynamics without explicit axiomatization — is sometimes preferred on the grounds that it avoids the circularity risks of the axiomatic method. The present paper shows that Phase Theory, by admitting multiple equivalent axiomatic presentations, achieves the robustness that critics of the axiomatic method seek: the theory's content is not trapped in any single presentation but is accessible from multiple independent starting points, exactly as constructive theories achieve their independence from any single calculational scheme.

9. Relation to Mathematical Precedents

The phenomenon of structural equivalence under reaxiomatization is well-established in mathematics and mathematical physics, and it is instructive to examine the precedents in detail, both to contextualize the present results and to draw lessons about the significance of equivalence for the standing of a foundational theory.

The oldest and most celebrated example is Euclidean plane geometry. Euclid's original axiomatization, as presented in the *Elements* [40], employed five postulates including the famous parallel postulate. This system, while historically transformative, was subsequently recognized to contain gaps — assumptions that Euclid made implicitly without stating them as axioms. Hilbert's rigorous reformulation of 1899 [1] corrected these gaps, providing a complete and independent set of axioms in terms of betweenness, congruence, and incidence. Birkhoff's reformulation of 1932 [2] took a different approach, using distance and angle as measured primitives and thereby reducing the axiom count to four. Tarski's first-order reformulation [3] — which admits complete decidability by Tarski's quantifier elimination theorem — is logically weaker in its language but captures exactly the same Euclidean theorems about geometric configurations. All three reformulations, despite their radical differences in primitive notions and logical strategy, generate precisely the same body of Euclidean theorems. The existence of this multiplicity of formulations is now understood as one of the strongest indicators of the richness and robustness of Euclidean geometry as a mathematical structure.

Classical mechanics provides the physical science precedent most directly relevant to Phase Theory. Newton's formulation [4] in terms of force, mass, and absolute time yields, for any mechanical system, a set of second-order differential equations — Newton's laws — that determine the system's trajectory given initial positions and velocities. Lagrange's formulation [5]

replaces this with a variational principle: the physical trajectory is that path in configuration space for which the action functional $S = \int L dt$ is stationary. Hamilton's formulation [6, 7] further transforms the description into one on phase space, where the canonical equations are first-order and the symplectic structure of the phase space is the central geometric object. For any fixed mechanical system, all three formulations yield identical trajectories: their empirical content is identical. Yet their conceptual structures are genuinely distinct, and each formulation offers advantages that the others do not — Lagrangian mechanics for constrained systems, Hamiltonian mechanics for quantization and perturbation theory, Newtonian mechanics for direct force analysis.

Thermodynamics presents a further instructive case. The second law of thermodynamics has been axiomatized in at least three distinct ways: the Kelvin-Planck formulation (no cyclic process can convert heat entirely into work), the Clausius formulation (heat cannot pass spontaneously from a cooler to a warmer body), and the Carathéodory formulation [41] (in any neighborhood of any thermodynamic state, there exist states inaccessible by adiabatic processes). These three formulations are logically equivalent — each implies the others — yet their conceptual starting points differ substantially. Carathéodory's formulation is particularly notable for its mathematical elegance: it expresses the second law in terms of the accessibility structure of state space, anticipating the differential-geometric approach to thermodynamics that has been developed in subsequent decades. Phase Theory, with the $\mathcal{A}_3$ formulation, stands in a relation to the other formulations that is analogous to Carathéodory's relation to Kelvin-Planck and Clausius: it foregrounds the structural and geometric character of the theory at the expense of intuitive simplicity, thereby making the mathematical tools most directly applicable.

The placement of Phase Theory within this tradition of mature foundational frameworks — as a theory that admits multiple equivalent presentations — is not merely an honor. It is a substantive epistemological achievement: it means that the theory's physical content has been shown to be independent of any particular mathematical or conceptual presentation, and that the choice of presentation can be made on grounds of convenience, elegance, or computational efficiency without affecting the physical conclusions.

10. The Irreducible Core

Having established that $\mathcal{A}_0$, $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ are structurally equivalent, we are in a position to identify and formally characterize the irreducible core of Phase Theory: the set of commitments that is necessary and sufficient for any axiom system to be a formulation of Phase Theory in the structural sense of this paper.

Definition 10.1 (Irreducible Core).

The *irreducible core* of Phase Theory is the maximal set of commitments that is shared by all axiom systems structurally equivalent to $\mathcal{A}_0$. A commitment is in the irreducible core if and only if its negation is incompatible with structural equivalence to $\mathcal{A}_0$.

Theorem 10.1 (Irreducible Core Theorem).

Across all axiom systems structurally equivalent to $\mathcal{A}_0$, the following five commitments are necessary and sufficient:

C1 (Phase Primacy): Phase is the sole primitive entity of physical reality.

C2 (Admissibility): All physically realizable configurations satisfy a global consistency condition governed by a functional $\mathcal{I}[\Phi] = 0$.

C3 (Admissibility-Preserving Evolution): Physical evolution is admissibility-preserving reconfiguration of phase, and the temporal ordering parameter is emergent from — not antecedent to — the evolution structure.

C4 (Structural Observability): Observables are structural features of phase — specifically, invariant or extremal functionals on the admissible configuration space.

C5 (Ontological Closure): No non-phase primitives are permitted at any stage of the theory.

Proof of Theorem 10.1.

Necessity: Each of C1 through C5 is present, in explicit or implicit form, in all four systems $\mathcal{A}_0$ through $\mathcal{A}_3$, as verified by inspection of the equivalence maps constructed in Sections 4 through 6. The negation of any one of C1 through C5 would violate at least one of the four conditions of Definition 3.1: negating C1 would violate condition (i) by permitting non-phase configurations; negating C2 would violate condition (i) by removing the admissibility selection; negating C3 would violate condition (ii) by permitting non-admissibility-preserving evolutions or introducing background time;

negating C4 would violate condition (iii) by altering the emergent structure; negating C5 would violate condition (iv) by permitting forbidden primitives. Therefore, each of C1 through C5 is necessary.

Sufficiency: Any axiom system satisfying C1 through C5 generates: the same admissible configuration space (by C1 and C2), the same admissible evolutions (by C3), the same observables and emergent structures (by C4), and the same exclusions (by C5). By Definition 3.1, it is therefore structurally equivalent to $\mathcal{A}_0$. Thus, C1 through C5 are sufficient.

■

Corollary 10.1.

Any axiom system satisfying C1 through C5 is a structurally equivalent formulation of Phase Theory. In particular, the three alternative systems $\mathcal{A}_1$, $\mathcal{A}_2$, and $\mathcal{A}_3$ are all equivalent formulations of Phase Theory in this sense, and no other set of commitments need be verified for equivalence beyond C1 through C5.

Remark 10.1 (Mutual Independence of C1–C5).

The five commitments C1 through C5 are mutually independent: no one of them can be derived from the remaining four alone. C1 is independent of C5 because one can conceive of a closed theory (satisfying C5) whose unique primitive is not phase but some other entity — a metric structure, a causal order, a spin foam. C2 is independent of C1 because one can have a phase-based ontology (satisfying C1) with local rather than global consistency conditions — a different constraint structure that would generate different physics. C3 is independent of C1 and C2 because one can have phase as the

unique primitive and a global admissibility condition, while still admitting background-dependent evolution — as in standard quantum field theory on a fixed curved spacetime, were it reformulated in phase-theoretic language. C4 is independent of C1 through C3 because the definition of observables is a further structural commitment not entailed by the existence of an admissible phase ontology with evolution. C5 is independent of C1 through C4 because one can satisfy all of the preceding commitments while retaining the option of introducing auxiliary fields or ghost degrees of freedom at an intermediate stage of calculation — a practice excluded by Closure but compatible with the other four commitments individually.

11. Summary of Results

The following is a concise inventory of the principal results established in this paper.

Alternative axiom systems constructed. Three alternative axiom systems have been introduced: $\mathcal{A}_1$ (Constraint-First Formulation), in which global consistency is the primary logical commitment and phase ontology is derived; $\mathcal{A}_2$ (Evolution-First Formulation), in which admissible change is the primary commitment and temporal ordering, phase ontology, and admissibility structure are all derived features; and $\mathcal{A}_3$ (Topology-Centered Formulation), in which topological phase reality is the primary commitment and the connection to topological quantum field theory is made mathematically manifest. Each system consists of six explicitly stated axioms in formal definition-style environments.

Equivalence maps proved. Six equivalence maps have been constructed: the forward and backward maps for each of the three pairs $(\mathcal{A}_0, \mathcal{A}_1)$, $(\mathcal{A}_0, \mathcal{A}_2)$, and $(\mathcal{A}_0, \mathcal{A}_3)$. Each map is an explicit axiom-by-axiom correspondence, and each

proof establishes all four conditions of Definition 3.1 in both directions. The Main Equivalence Theorem (Theorem 3.1) — $\mathcal{A}_0 \cong \mathcal{A}_1 \cong \mathcal{A}_2 \cong \mathcal{A}_3$ — is thereby established, with structural equivalence in the sense of Definition 3.1.

Invariant core isolated. The Irreducible Core Theorem (Theorem 10.1) identifies five commitments — C1 through C5 — that are both necessary and sufficient for structural equivalence to $\mathcal{A}_0$. These five commitments are shown to be mutually independent (Remark 10.1). Corollary 10.1 provides a practical criterion: any axiom system satisfying C1 through C5 is a formulation of Phase Theory, and no additional verification is required.

Physical consequences for the standing of Phase Theory. Theorem 7.1 establishes that all falsifiable physical predictions of Phase Theory are presentation-invariant. The critique that Phase Theory's conclusions are artifacts of its particular axiomatic presentation is structurally refuted. Phase Theory joins the class of mature foundational frameworks — alongside Euclidean geometry, classical mechanics, and thermodynamics — that admit multiple structurally equivalent presentations, a property recognized across the history of science as a hallmark of theoretical depth and robustness.

12. Conclusion

The central result of this paper is that Phase Theory is not axiom-fragile. The physical content of the theory — its admissible configurations, its evolutions, its emergent structures, its observables, and its falsifiable predictions — is invariant under the choice of axiom presentation. Three structurally distinct alternative formulations have been constructed and proved equivalent to the canonical system, and the irreducible core of the theory has been identified and formally characterized. Phase Theory is established, by the results of this paper, as a theory whose content is structural rather than definitional.

The philosophical significance of this result deserves emphasis. A physical theory whose conclusions depend essentially on its particular axiomatic presentation is, in a precise sense, not making claims about the physical world: it is making claims about a particular way of organizing language about the physical world. The demonstration that Phase Theory's conclusions survive radical reaxiomatization — that they emerge from constraint-first, evolution-first, and topology-centered starting points with equal force — establishes that the theory is making genuine structural claims about physical reality. The phase-theoretic picture of the world as a space of admissible topological configurations, evolving continuously through admissibility-preserving trajectories, with observables determined by topological invariants and extremal functionals, is a picture of the world — not a picture of a formalism.

The practical implications for the subsequent development of the series are substantial. Papers 22 and beyond will exploit the three alternative formulations selectively, deploying whichever presentation is most natural for the problem at hand. The $\mathcal{A}_3$ (topology-centered) formulation will serve as the primary framework for particle classification, where the classification of topological defect structures by their admissibility-compatible topological invariants provides the most direct mathematical route to particle quantum numbers. The $\mathcal{A}_2$ (evolution-first) formulation will serve as the natural framework for cosmological applications, where the emergence of cosmic time from an initial phase configuration — rather than its assumption as a background parameter — is the technically and conceptually central question. The $\mathcal{A}_1$ (constraint-first) formulation will be exploited in the analysis of the relationship between Phase Theory and canonical approaches to quantum gravity, where the physical Hilbert space is defined as the kernel of a constraint operator and the structural parallels with $\mathcal{A}_1$ are most direct.

The ultimate significance of the results established here lies in their bearing on what it means for a physical theory to be genuine. A theory is genuine — rather than merely a sophisticated system of conventions — if its content cannot be dissolved by reformulation, if its predictions survive the exchange of primitive and derived status, if its structure is accessible from multiple independent starting points. Phase Theory, as demonstrated in this paper, satisfies all these conditions. The theory's commitment to phase as the irreducible substrate of physical reality, to admissibility as the selector of the physical world from the space of all possibilities, and to emergent spacetime and matter as consequences rather than inputs, constitutes a genuine structural claim about the architecture of nature — a claim that stands independent of the particular axiomatic form in which it is expressed.

¹ The notion of "baking in conclusions" is sometimes called the problem of *analytic entailment* in philosophy of science. See Hempel [34] for a classical discussion and Quine [42] for the broader context of the analytic-synthetic distinction's instability in physical theories.

² The Carathéodory axiomatization of thermodynamics [41] is the most direct historical precedent for the $\mathcal{A}_1$ formulation in spirit: both begin from accessibility or admissibility conditions rather than from explicit ontological declarations.

³ The equivalence of the Kelvin-Planck and Clausius formulations of the second law is a standard exercise in thermodynamics textbooks, but the proof is non-trivial and requires the concept of a Carnot cycle as an intermediary. The analogy with the equivalence maps of Section 3 is instructive.

⁴ The treatment of observables in $\mathcal{A}_3$ as topological invariants is closely related to, but not identical with, the definition of observables in a TQFT. In an Atiyah-type TQFT [18], observables are defined by the functorial structure of the theory; in $\mathcal{A}_3$, they are defined by the admissibility-preserving invariance condition. The relationship between these two approaches is explored in Paper 14 of the Phase Theory series.

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